

MELITA, MB, CanadaWMO: **714470**Lat: **49.2812N**Lon: **100.9862W**Elev: **446**StdP: **96.08**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.0	-29.3	-35.3	0.1	-31.9	-32.5	0.2	-29.2	12.8	-12.2	11.5	-11.0	2.8	260	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	30.2	20.3	28.6	19.6	27.0	18.8	22.7	27.5	21.4	26.3	20.2	24.8	4.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.7	25.3	19.7	15.3	24.1	18.5	14.1	22.9	69.5	27.5	64.7	26.2	60.1	24.7	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.9	8.8	DB	-35.8	34.0	2.8	1.9	-37.8	35.3	-39.5	36.5	-41.0	37.6	-43.0	39.0	(4)
(5)			WB	-35.9	24.9	2.8	1.8	-37.9	26.2	-39.6	27.2	-41.1	28.2	-43.1	29.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-14.8	-12.9	-5.5	4.3	11.2	16.6	19.3	18.5	13.4	5.3	-4.0	-11.8	(6)
(7)		DBStd	13.59	8.53	7.97	7.83	5.95	4.71	3.62	3.00	3.45	4.53	5.20	6.71	7.79	(7)
(8)		HDD10.0	3391	767	641	480	184	42	2	0	0	18	162	419	676	(8)
(9)		HDD18.3	5587	1026	874	738	421	225	73	24	41	158	404	668	934	(9)
(10)		CDD10.0	981	0	0	1	14	79	200	289	264	119	16	0	0	(10)
(11)		CDD18.3	135	0	0	0	0	4	22	55	46	9	0	0	0	(11)
(12)		CDH23.3	1568	0	0	0	6	81	236	529	553	155	9	0	0	(12)
(13)		CDH26.7	415	0	0	0	1	19	50	127	177	39	2	0	0	(13)
(14)	Wind	WSAvg	4.2	4.1	4.2	4.4	4.7	4.8	4.0	3.5	3.6	4.0	4.4	4.2	4.1	(14)
(15)	Precipitation	PrecAvg	451	18	15	20	30	59	84	68	58	41	29	18	20	(15)
(16)		PrecMax	576	42	48	64	70	185	236	142	186	91	80	73	41	(16)
(17)		PrecMin	277	7	6	1	9	12	28	14	19	12	0	2	2	(17)
(18)		PrecStd	93	9	9	13	15	42	47	29	35	21	20	16	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	5.0	15.6	23.7	29.3	30.7	31.4	33.5	30.2	24.1	14.1	6.0	(19)
(20)			MCWB	1.8	2.8	9.6	12.8	16.2	20.7	22.3	20.5	18.7	14.0	8.4	3.1	(20)
(21)		2%	DB	1.9	2.3	10.2	19.9	25.4	27.9	29.7	30.6	27.4	19.6	10.1	2.7	(21)
(22)			MCWB	0.7	1.0	6.3	10.9	14.3	18.3	21.6	19.9	17.4	11.8	5.7	1.1	(22)
(23)		5%	DB	-0.4	0.5	6.1	17.3	22.7	26.0	28.1	28.6	24.8	16.5	7.0	0.5	(23)
(24)			MCWB	-1.3	-0.2	3.9	9.3	13.1	18.0	21.1	19.3	16.3	10.3	4.0	-0.6	(24)
(25)		10%	DB	-2.6	-1.5	3.5	14.1	20.3	24.2	26.4	26.5	21.9	13.4	4.3	-1.6	(25)
(26)			MCWB	-3.2	-2.1	2.1	8.1	12.1	17.1	19.7	18.4	14.9	8.9	2.4	-2.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	3.2	9.9	13.8	18.2	23.7	25.2	23.9	21.4	15.1	8.9	3.4	(27)
(28)			MCDWB	3.5	4.7	14.6	20.3	24.9	28.2	28.5	29.7	26.5	21.5	13.4	5.6	(28)
(29)		2%	WB	0.6	1.1	6.4	11.7	16.1	21.2	23.1	21.9	19.0	12.9	6.3	1.2	(29)
(30)			MCDWB	1.8	2.1	10.1	18.4	21.7	25.1	27.5	27.6	24.6	17.9	9.5	2.5	(30)
(31)		5%	WB	-1.2	-0.1	4.0	10.0	14.6	19.2	21.9	20.6	17.4	11.1	4.2	-0.5	(31)
(32)			MCDWB	-0.4	0.6	6.1	15.7	20.5	23.7	26.7	26.1	23.0	15.4	6.6	0.5	(32)
(33)		10%	WB	-3.2	-2.1	2.2	8.5	13.3	17.9	20.7	19.4	15.8	9.3	2.5	-2.3	(33)
(34)			MCDWB	-2.6	-1.5	3.4	13.6	18.4	22.4	25.0	24.7	20.6	13.3	4.3	-1.5	(34)
(35)	Mean Daily Temperature Range	MDBR		9.7	10.0	9.5	11.7	12.6	11.2	11.9	13.4	12.9	11.1	9.2	8.9	(35)
(36)		5% DB	MCDBR	10.4	9.5	12.0	16.7	16.6	14.1	14.1	16.5	17.5	16.6	13.3	10.7	(36)
(37)			MCWBR	9.5	8.3	8.5	8.7	7.8	7.1	7.3	7.7	8.8	9.5	9.2	9.3	(37)
(38)			MCDBR	10.6	9.1	11.4	15.3	13.5	11.8	12.8	13.6	14.9	15.3	12.6	10.8	(38)
(39)		5% WB	MCWBR	9.6	8.0	8.3	8.8	7.2	7.0	7.4	7.3	8.3	9.0	9.0	9.4	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.253	0.257	0.304	0.335	0.350	0.365	0.379	0.387	0.350	0.295	0.263	0.238	(40)
(41)		taud		2.332	2.348	2.377	2.385	2.393	2.395	2.370	2.326	2.402	2.533	2.489	2.410	(41)
(42)		Ebn at Noon		818	908	912	913	909	894	874	848	848	856	812	798	(42)
(43)		Edh at Noon		66	84	98	109	113	114	115	115	96	70	57	53	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.50	2.65	3.90	4.99	5.78	6.09	6.52	5.29	3.91	2.40	1.39	1.07	(44)
(45)		RadStd		0.12	0.23	0.30	0.36	0.40	0.42	0.29	0.40	0.31	0.28	0.19	0.11	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.56	N/A	N/A	N/A	N/A

Nomenclature: See separate page